Lecture 1: Course introduction

TTIC 31220: Unsupervised Learning and Data Analysis

Instructor: Karen Livescu

TAs: Chung-Ming Chien, Ju-Chieh Chou, Nirmit Joshi

TTI-Chicago

March 23, 2026

Plan for today

- Logistics
- Course introduction
- Supervised learning review

Logistics

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General: This is a graduate-level class

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Prerequisites:

- Probability, linear algebra, TTIC 31020 or equivalent, comfort with programming (we will use Python)
- Some background reading and exercises ("HW0") available on canvas
- Consent requests: Submit online; mention your prereqs/equivalents + grades
- Class is currently full; there is a waitlist

Logistics

Office hours:

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Tutorials (complementary to lectures):

- Wednesdays 12:30-1:20pm in room 529
- Tutorial 1 this week: Review of prerequisites
- Tutorials 2-4: Kernels, neural network nuts & bolts

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Course staff email address: unsup-learning-staff@ttic.edu

Grading, assignments

- 3-4 homework assignments (20%)
- 3 (brief) quizzes (25%)
- Term project (45%)
- Effort/participation (10%)

Project

- Ideally (possibly only) to be done with a partner
- We will provide suggested topics
- Can be analysis of some published work, analysis of a data set, or implementation of a new research idea
- Includes proposal, brief progress report, poster presentation, short conference paper-style final report

Readings

- No required textbooks
- Books that we will draw on (PDFs available via canvas site):
  - C. Bishop, *Pattern Recognition and Machine Learning*
  - K. P. Murphy, *Machine Learning: A Probabilistic Perspective*
- Additional readings will be provided as needed

Collaboration policy

- You are encouraged to discuss homework and any course material with others
- Submitted material (writeups, code) must be produced on your own
- Don't submit anyone else's words (including an LLM's) as your own

LLM use policy

- Use LLMs like you would use the course staff: Do not ask them to write down solutions for you, but do ask them to clarify points of confusion, explain some concept, etc.

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- Also use LLMs (or any other tool) to proofread your writing for grammar/typos.

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- Submitted solution must be your own words.
- If you used LLMs, describe how you did so and what you found most useful!

Other notes

This is a PhD-level class

- You will be expected to show some independence and fill in steps on your own (sometimes via the homework)
- We will be using a mix of readings from textbooks and the research literature, with different notations and occasionally terminology

Questions?

Course introduction

Types of learning

Learning = "using past experience to improve future outcomes"

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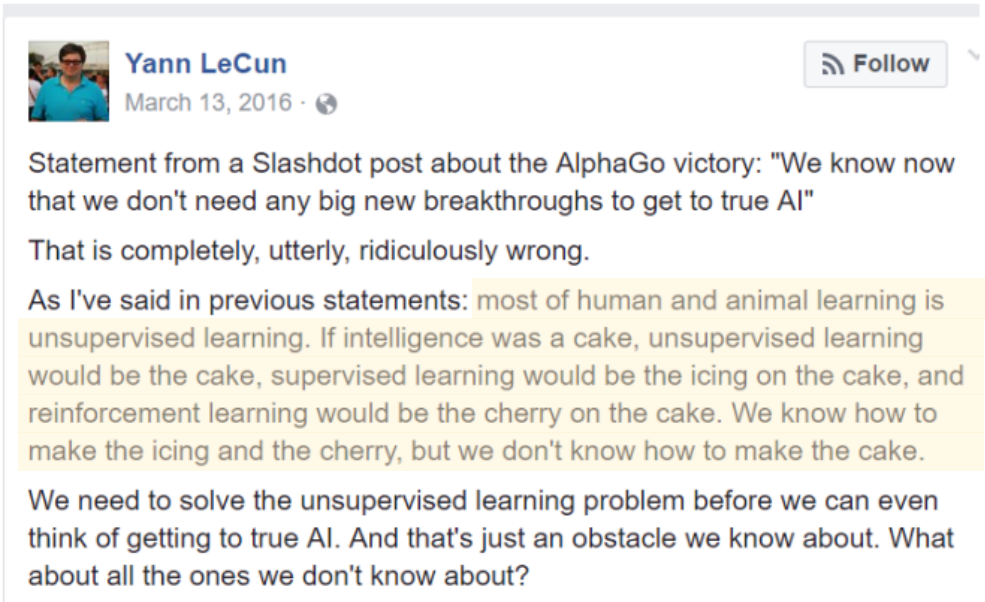
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- **Reinforcement learning** (e.g., TTIC 31170): learning to take actions based on rewards for past actions
- (**Game theoretic/multi-agent learning?**: like reinforcement learning, but the learner is an agent among multiple interacting agents)

Why unsupervised learning?





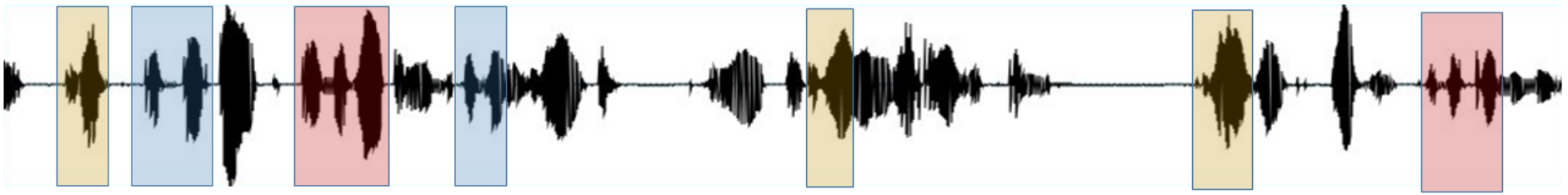
Unsupervised learning applications: Counting objects



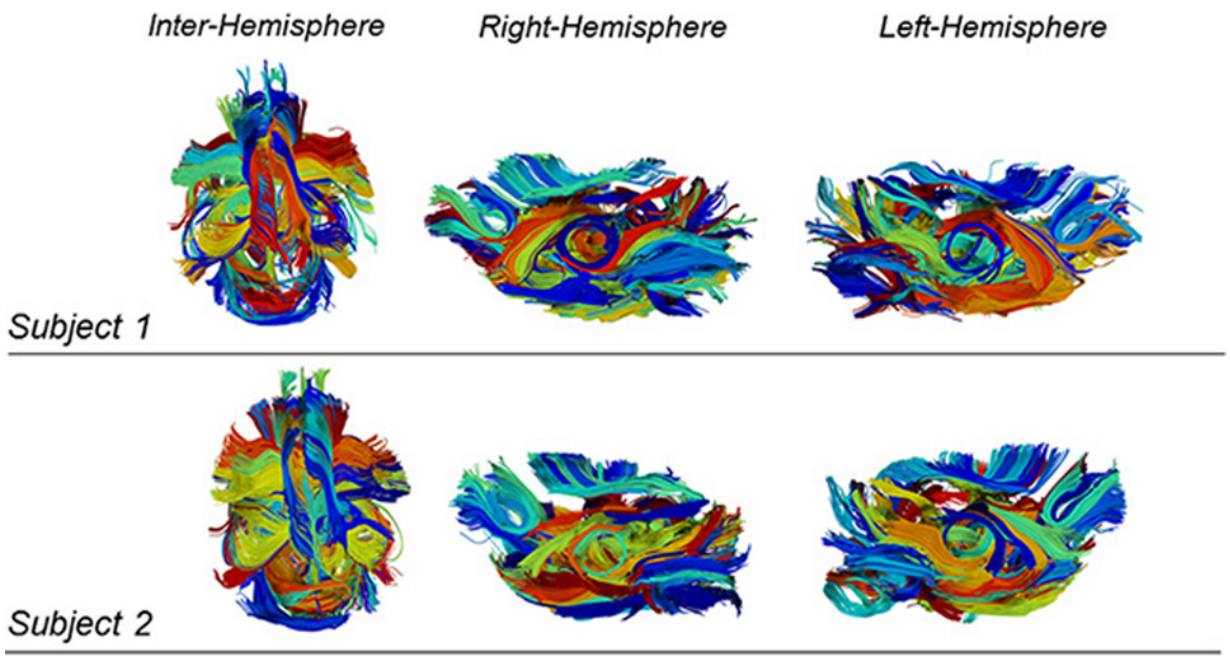
Unsupervised learning applications: Discovering topic hierarchies

Creationism; War; Afrocentrism; Wahhabism; Harvey Milk; Saunders Lewis; Jean-Jacques Rousseau; William Joice; Opera; Idi Amin; John Donne; History of the telescope; Asceticism; Afterlife; Information; Satire; Notary public; Romantic nationalism; Thomas Aquinas; List of Estonians; List of fantasy authors; List of Germans; Vatican City; Torah; Old Catholic Church; Apostles' Creed; Mahmoud Abbas; Boris Yeltsin; Ulysses S. Grant; William Jennings Bryan; Pipe organ; George IV of the UK; George V of the UK; Emperor Taizong of Tang; David I of Scotland	Hanoi Hilton; USS Hornet; Battle of Peleliu; House (astrology); Hua Mulan; 1972 in sports; Moselle; Duke of York; Spam (Monty Python); History of Vanuatu; Liliopsida; 1908 Summer Olympics; Lucy Liu; Celia Cruz; National Cartoonists Society; Balearic Islands; SCSI; Asexuality; Copyright infringement of software; Lamb of God; Calendar date; Lapland War; Totalitarianism; History of Sweden; European Youth Parliament; Roman mythology; Kingdom of Israel; List of railway companies; Maundy money	Florence Nightingale; Texaco; Poltergeist; Isadora Duncan; John Allan Muhammad; 1935 in music; 1934 in music; 1939 in music; 1921 in music; Star Trek; 1776 (musical); The Godfather Part II; Dial M for Murder; Jimmy Durante; 2006 Commonwealth Games; 1968 in music; 1977 in music; Comet Hale-Bopp; High Speech Rail; Steve Wozniak; James Carville; David Copperfield; Lynyrd Skynyrd; Poison (band); LL Cool J; RZA; Stand; Bob Dole; Sergei Prokofiev; Peter Hain; Ravana; Michael Porillo; George Pickett	Roswell, SD; LaFayette, KY; Old Ripley, IL; Hainesville, IL; Belleville, WI; Bethany, IL; South Point, Ohio; Ashwaubenon, WI; North Miami, FL; Davenport, IA; Spartanburg, SC; Hurricane Lili; Scranton, PA; Myopia; Influenza; Motion picture rating system; FEMA; Vineland, NJ; Fair Lawn, NJ; Medford, NJ; Palisades Park, NJ; South River, NJ; Gillespie County, TX; Hidalgo County, NM; Finney County, KS; Onslow County; NC; Brady Township, PA; Argentine Township, MI; Augusta Gharter Township, MI	Natural number; Linear subspace; Cauchy distribution; Wrench; List of matrices; Hydrogenation; Alpha helix; Heat pump; Soil; Bicarbonate; Campfire; Thermal depolymerization; Cardiac arrest; Neuron; Transformer; Hepatocellular carcinoma; Yellow fever; Isopropyl alcohol; Gorilla; Make (software); Ostinato; Control flow; PHP; HTML; Objective-C; Vienna Development Method; Comparison of Java C++; Dot product; Elliptic integral; Catalan number; Expected value; Coenzyme; Neutron; History of geodesy; Orbital period

Unsupervised learning applications: Discovering units of speech in a language

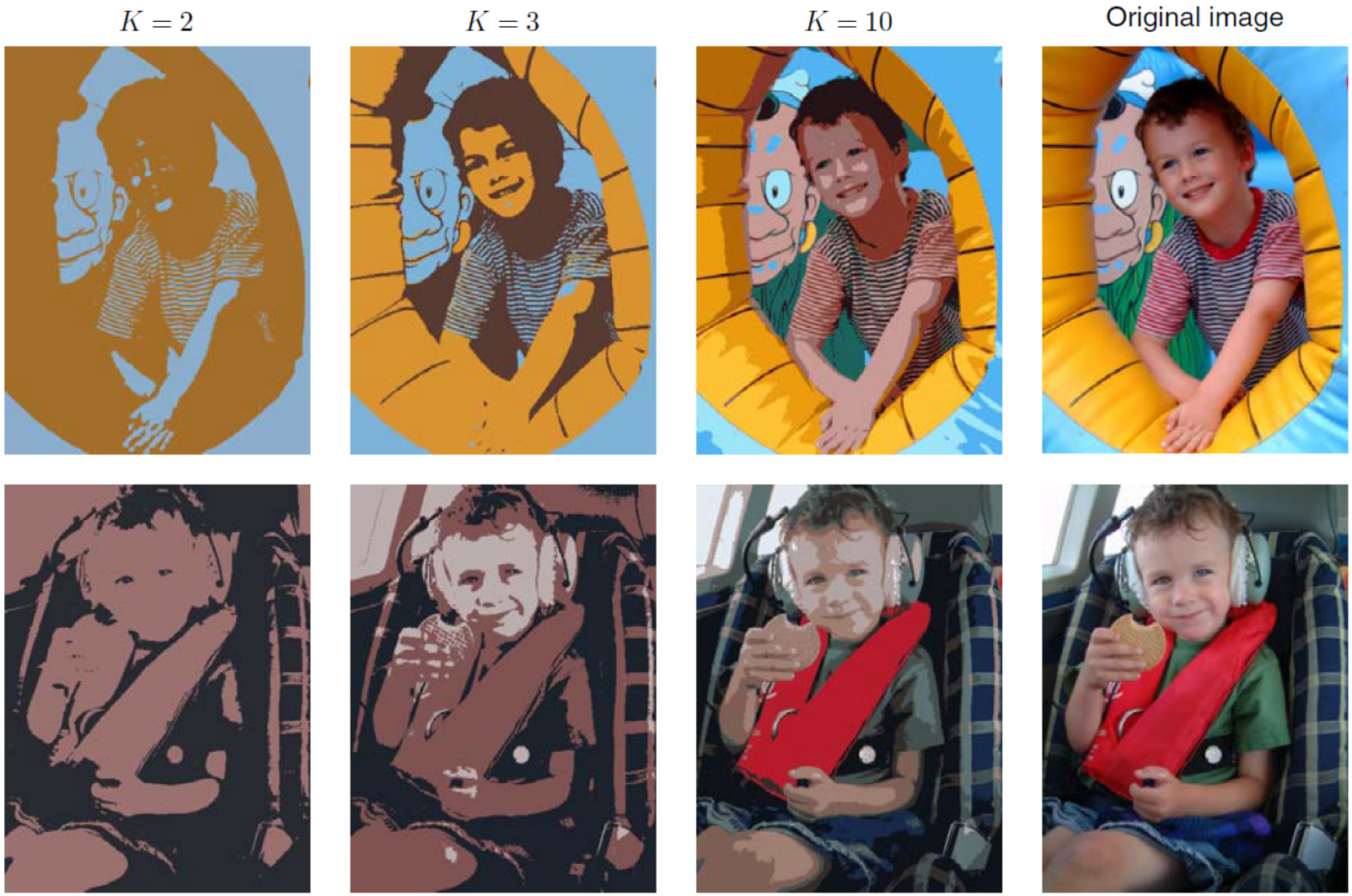


Unsupervised learning applications: Determining "modules" of brain activity



[Dodero *et al.* Front. Neuroinform. 2015]

Unsupervised learning applications: Image compression



Unsupervised learning applications: Generation



after 5 epochs



after 100 epochs





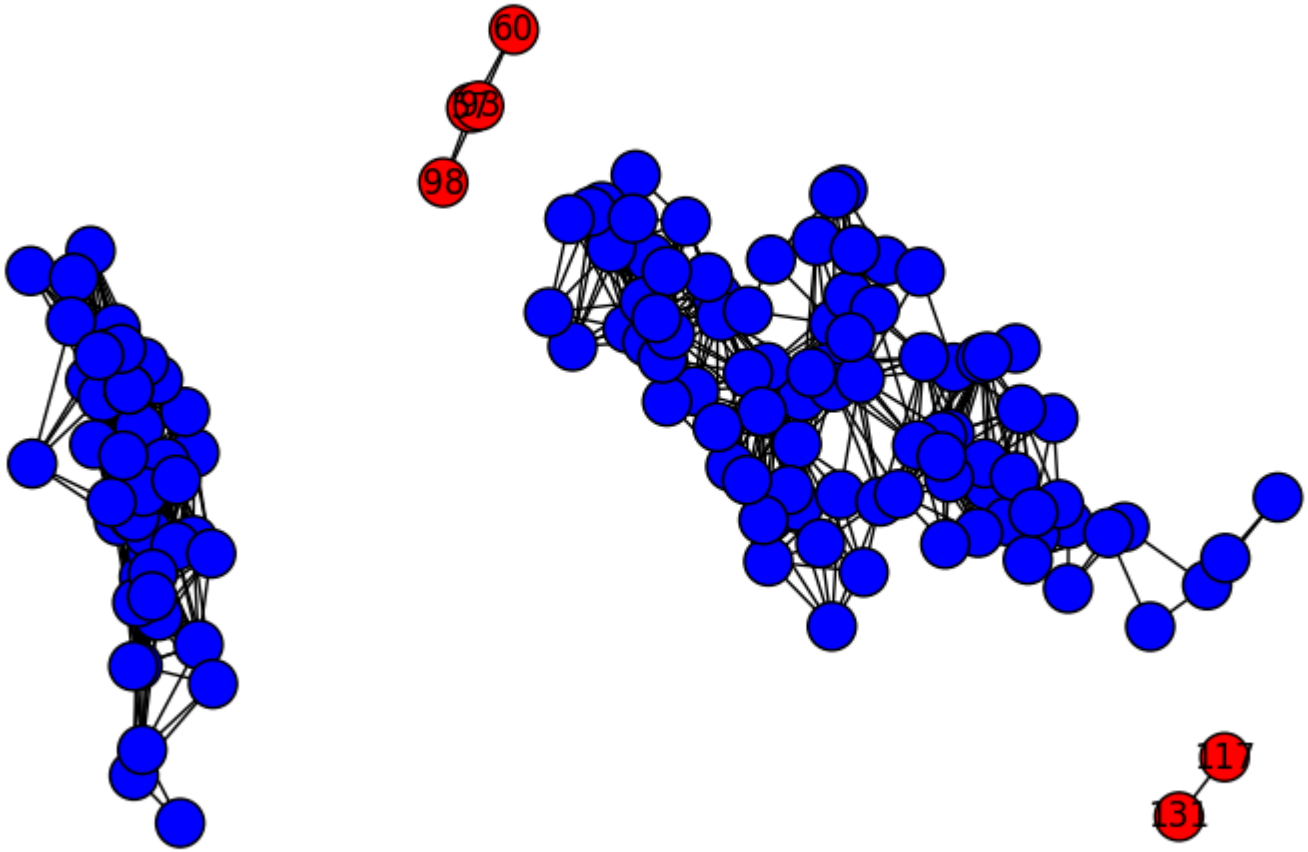
Unsupervised learning applications: Generation

John St Clair Etouffee, world-renowned art expert, connoisseur, aesthete, art historian, and lecturer at the University of Oxford, is walking around the Louvre, savoring the exquisite artwork of all ages assembled before him. The Louvre is closed, allowing him free access to all the paintings and silence in which to meditate them. He stops in front of La Gioconda, Leonardo da Vinci's world-famous Mona Lisa. After a time, he begins to speak and describe the qualia, the frisson, the nigh-indescribable experience of the subtle swirls & *sfumato* of the painting:

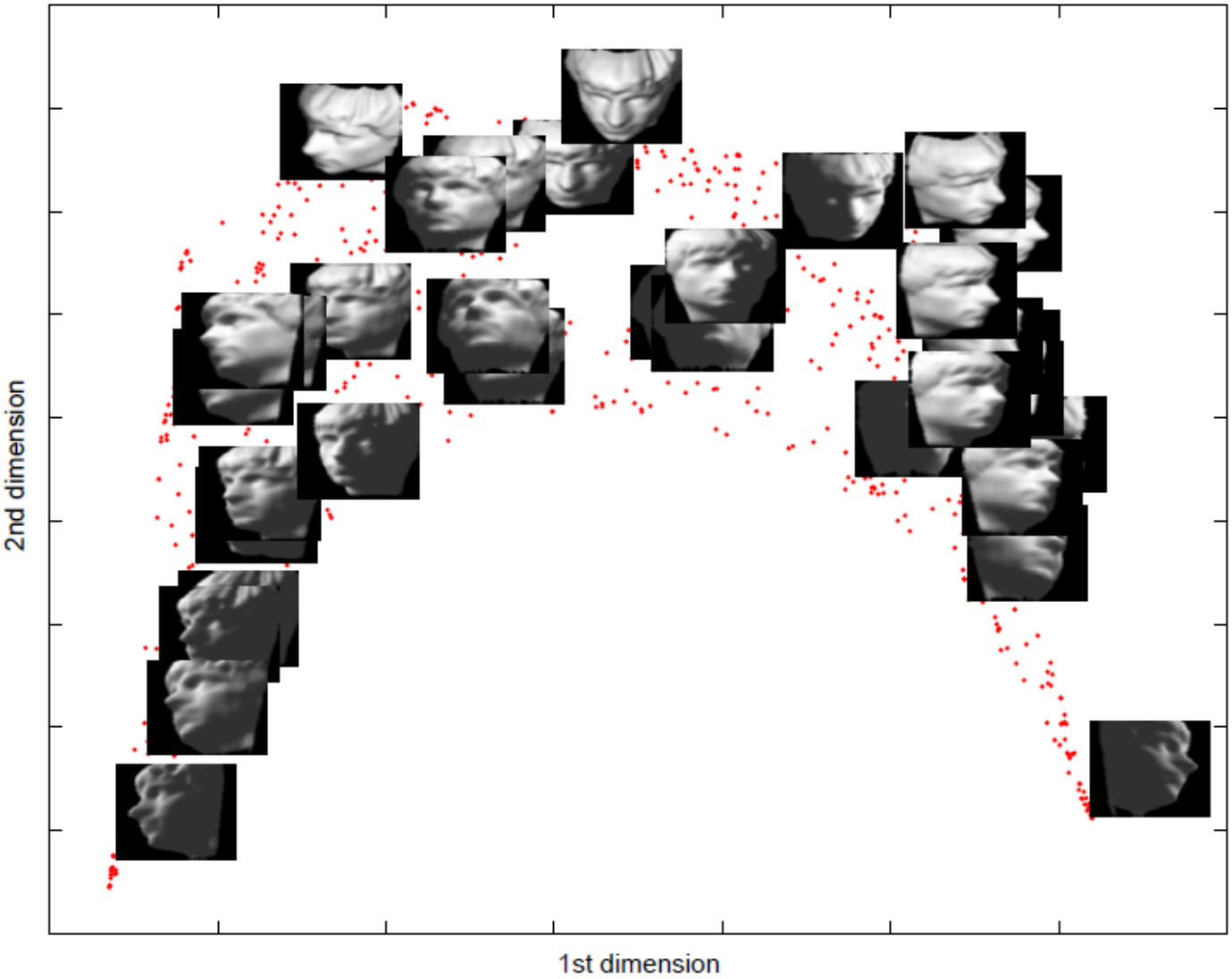
"The sun is now barely a foot or two below the horizon and with every second, the golden tone of the painting deepens. Everything melts in it. I, who am having this experience, appear to you in a little corner of the painting and I look at you through the painting and our eyes meet. Our arms, our shoulders, our heads, our hands, our necks, our mouths, our lips, our eyes, our fingers and the tips of our toes—all this transforms itself into bright, beauteous colour. Everything becomes alive, all there is left in me is love and appreciation. It makes me want to shout at you "Don't leave!" but you don't hear me and I remain silent. La Gioconda looks at me and I at La Gioconda, the same as before, but now it's like being in two places at the same time and my feet have no feeling."

St. Clair Etouffee leaves the painting and enters the Hall of the Kings, and of Kings. He moves on to the Indian Wars, passes the Saloon d'Orléans, the King of Britain with his Tudor and Stuart wives, and is reminded of what St Clair had read a lifetime earlier at the age of twelve in a book about medieval paintings

Unsupervised learning applications: Outlier/anomaly detection



Unsupervised learning applications: Representation learning for visualization



Unsupervised learning applications: Representation learning for visualization

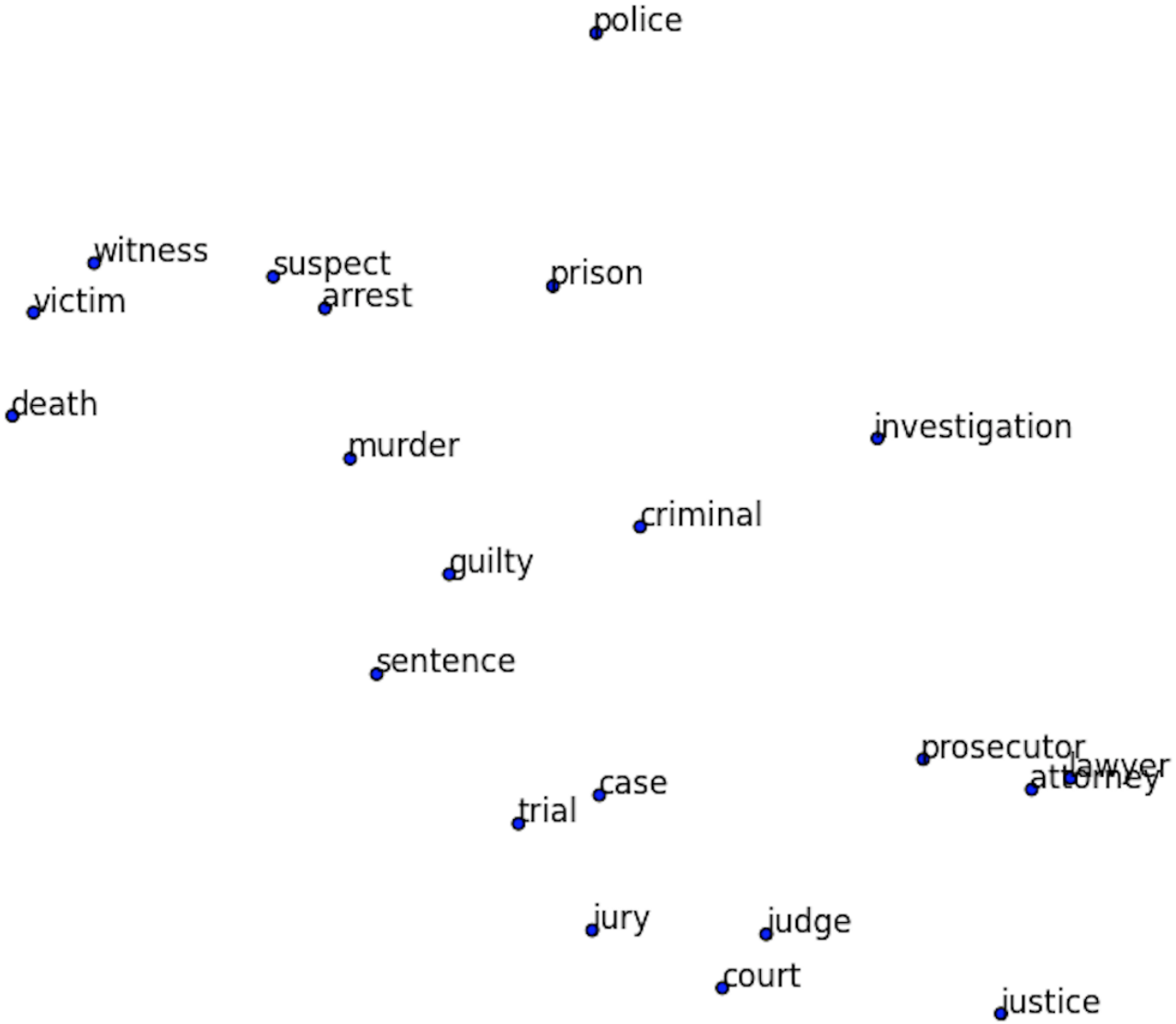


Unsupervised learning applications: Learning representations for better supervised learning

Typical example: Lots of unlabeled data X_U, a little labeled data X_L

- Learn representation from unlabeled data
- Use learned representation to improve performance of supervised learner on labeled data

Unsupervised learning applications: Word embeddings



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  - For compression
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Note: The boundaries between these classes is fuzzy!

Evaluating unsupervised methods

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It's hard!

- Qualitative evaluation
- Evaluation on supervised downstream task using the learned representation/clustering/density
- "Intrinsic" evaluation measures, e.g. cluster purity, degree of compression, distribution similarity...

Combining supervised and unsupervised learning

Unsupervised learning as a preprocessing step for supervised learning

- Learn features from a lot of unsupervised data
- Use them to learn a classifier/regressor using a small amount of supervised data

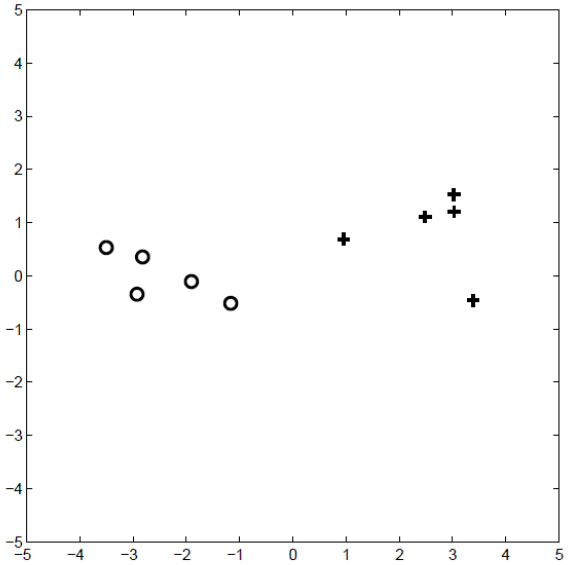
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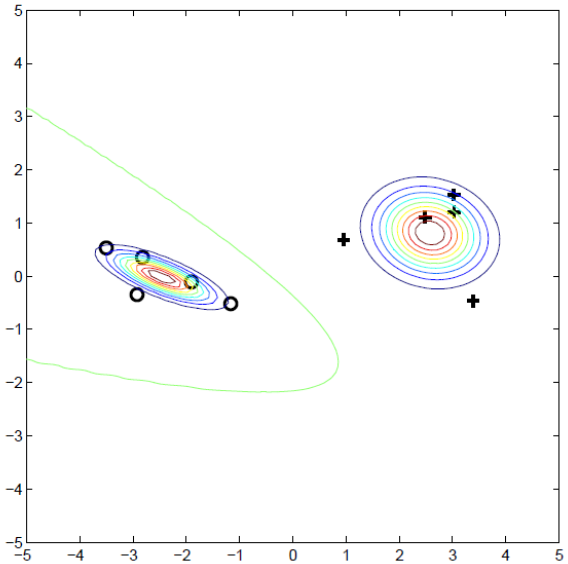
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Direct combination of unlabeled and labeled data (semi-supervised learning)

Semi-supervised learning

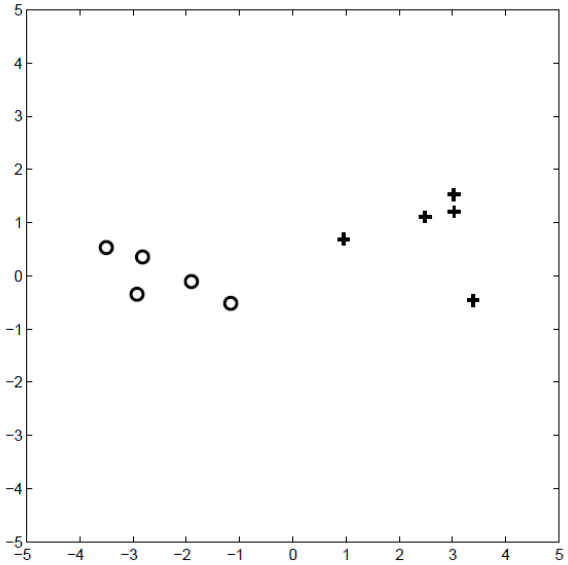


(a) labeled data

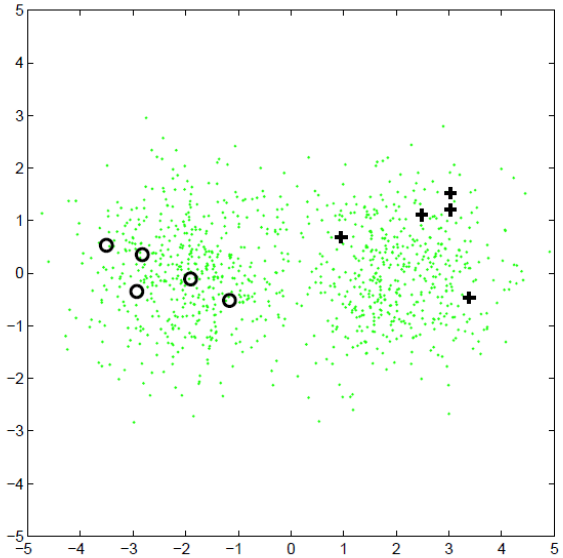


(c) model learned from labeled data

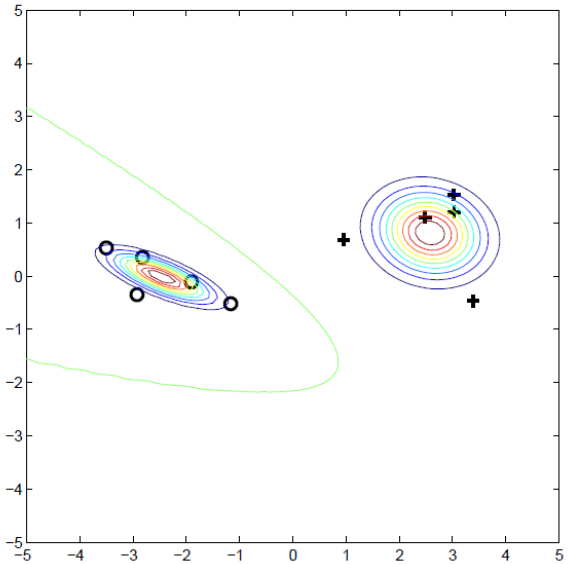
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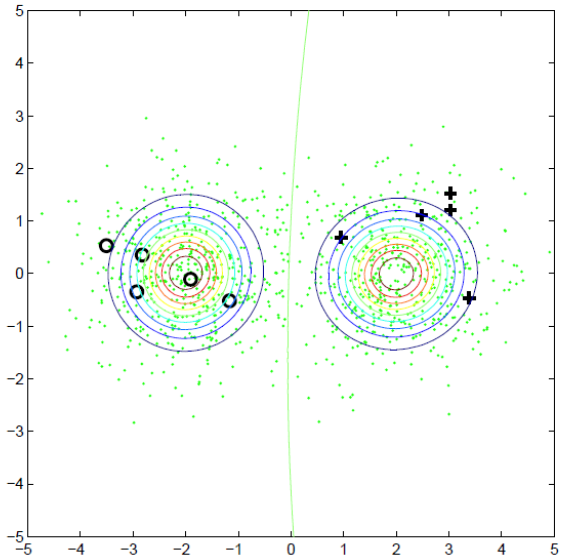
(a) labeled data



(b) labeled and unlabeled data (small dots)



(c) model learned from labeled data



(d) model learned from labeled and unlabeled data

A philosophical note

Some questions about terminology

- The boundary between supervised and unsupervised learning is fuzzy
- Maybe there is no such thing as unsupervised, only "differently supervised"
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- We will focus on a set of **important methods** that fall **outside the typical supervised learning setting**

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On a personal note

- My research often combines unlabeled and labeled data, rarely using a "pure" unsupervised learning approach
- But it's useful to think about unsupervised and supervised learning in isolation, and then combine them as needed

More philosophy

Some unsupervised learning doesn't look like learning at all

- E.g., visualization
- Recall: **Learning** = "using past experience to improve future outcomes"

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Theory vs. practice in unsupervised learning

- Theory is especially far from reality for unsupervised learning
- But theoretical results can give insight and intuition, and will be discussed in such cases
- Overall, you will find more practice than theory in this course

(Rough, tentative) lecture plan

- Introduction (1)
- Dimensionality reduction/representation learning (3-4)
- Clustering, topic modeling (3)
- Density estimation: mixtures, expectation-maximization, sequence models (3)
- Self-supervised learning (2)
- Semi-supervised learning (1)
- Applications, other advanced topics (3-5)

Supervised learning review

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  - Training data: pairs $(\mathbf{x}_i, y_i)$ labeled by a "teacher"
  - Function (hypothesis) class F, usually parametric $\{f(\mathbf{x}; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
  - Loss function ℓ penalizing for predicting $\widehat{y}$ when ground truth is y

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$$\boldsymbol{\theta}^* = \operatorname*{argmin}_{\boldsymbol{\theta}} \sum_i \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i)$$

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- **Note**: ℓ is usually a *surrograte loss*, related to the real task but not exactly the same as the task loss (that we really care about)

Loss: empirical and expected

- **Fundamental assumption**: example $\mathbf{x}$/label y are drawn i.i.d. from a joint probability distribution $p(\mathbf{x}, y)$ (*)
- Same (unknown!) distribution for all pairs $(\mathbf{x}, y)$ in both training and test data

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$$\frac{1}{N} \sum_{i=1}^{N} \ell \left(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i \right)$$

- The ultimate goal is to minimize the *expected loss*, also known as *risk*:

$$E_{(\mathbf{x},y)\sim p(\mathbf{x},y)} \left[\ell \left(f(\mathbf{x}; \boldsymbol{\theta}), y \right) \right]$$

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Loss: empirical and expected

- **Empirical loss**:

$$\frac{1}{N}\sum_{i=1}^{N}\ell\left(f(\mathbf{x}_i,\boldsymbol{\theta}),y_i\right)$$

- **Expected loss**:

$$E_{(\mathbf{x},y)\sim p(\mathbf{x},y)}\left[\ell\left(f(\mathbf{x},\boldsymbol{\theta}),y\right)\right]$$

- **Empirical risk minimization (ERM) approach**: to the extent that the training set is a representative of the underlying distribution $p(\mathbf{x},y)$, the empirical loss serves as a proxy for the risk (expected loss)
- Technically: estimate $p(\mathbf{x},y)$ by the *empirical distribution* of the data

Review: supervised learning

- **Task**: build a mapping from input $\mathcal{X}$ to output $\mathcal{Y}$, given training set $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, with $\mathbf{x}_i \in \mathcal{X}$, $y_i \in \mathcal{Y}$.

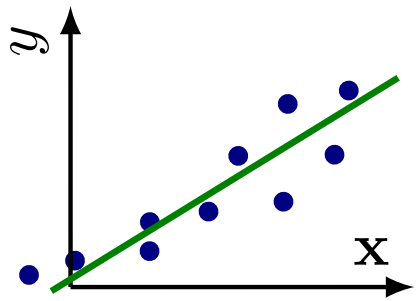
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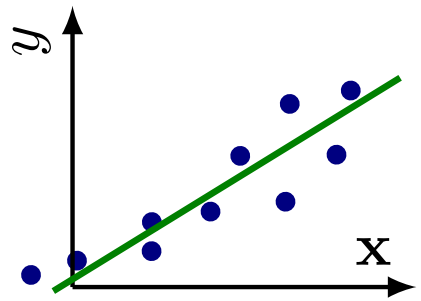
- Regression: $\mathcal{Y} = \mathbb{R}$
  learn a function $y = f(\mathbf{x}; \boldsymbol{\theta})$



Review: supervised learning

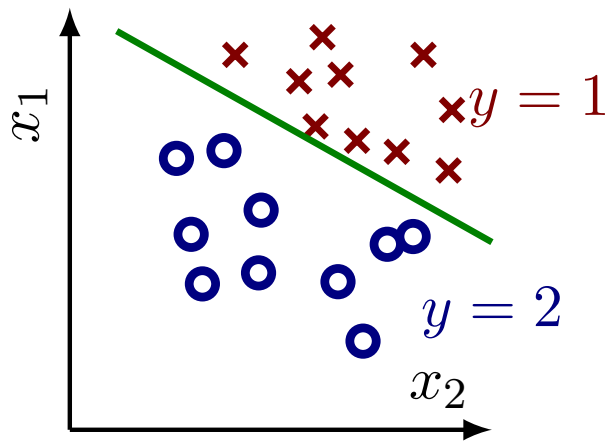
- **Task**: build a mapping from input $\mathcal{X}$ to output $\mathcal{Y}$, given training set $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, with $\mathbf{x}_i \in \mathcal{X}$, $y_i \in \mathcal{Y}$.
- **Main types of supervised learning**

- Regression: $\mathcal{Y} = \mathbb{R}$
  learn a function $y = f(\mathbf{x}; \boldsymbol{\theta})$



- Classification: $\mathcal{Y} = \{1, \ldots, C\}$
  learn a separator between classes, usually based on a *score function*

$$y = \operatorname*{argmax}_c f_c(\mathbf{x}; \boldsymbol{\theta})$$



Supervised learning I: Regression

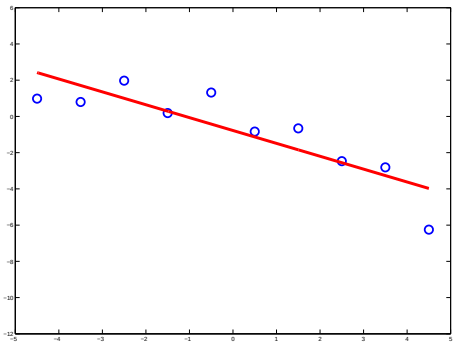
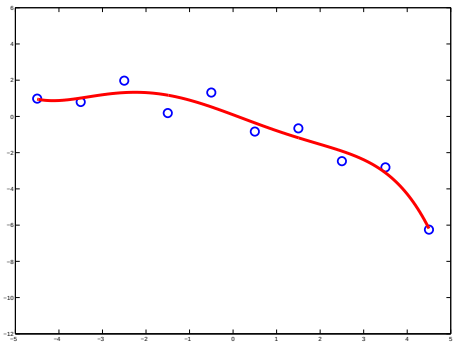
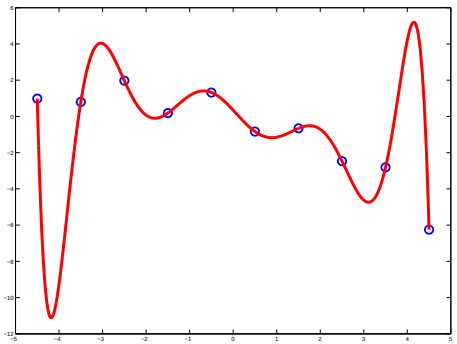
Typical approach:

- Design a *feature mapping* / representation $\boldsymbol{\phi} : \mathcal{X} \to \mathbb{R}^d$
- Minimize squared loss for a linear model $y = \mathbf{w} \cdot \boldsymbol{\phi}(\mathbf{x})$
- Model complexity driven by choice of $\boldsymbol{\phi}$

Supervised learning I: Regression

Typical approach:

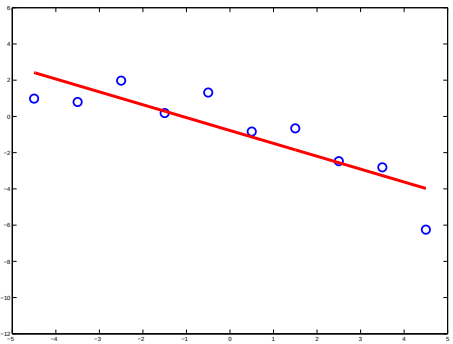
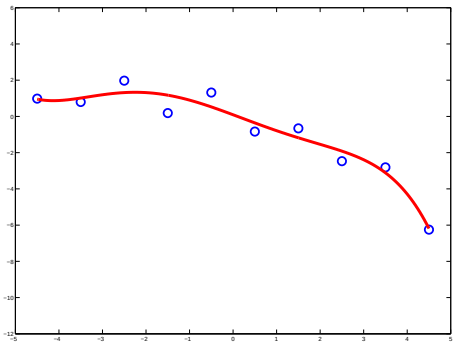
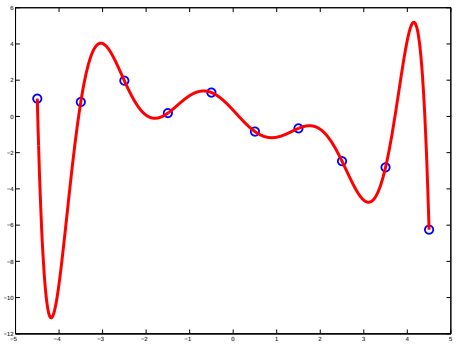
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- Example: Fitting polynomial models of order 1,5,10:

Supervised learning I: Regression

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- Model complexity driven by choice of ϕ
- Example: Fitting polynomial models of order 1,5,10:

- Not complex enough: **underfit**
  can sometimes tell by looking at training error
- Too complex: **overfit**
  will not see on training set – need to look at validation set

Model complexity and overfitting

- **Generalization** is real learning ability: how well we use at test time what we have learned during training
- **Capacity** is a measure of how well the model can fit the training set
- **Overfitting** typically corresponds to high capacity, poor generalization
- We often consider nested model families (linear $\rightarrow$ quadratic $\rightarrow \ldots$)

Regularization

- **Intuition**: penalize more complex models to avoid overfitting

Regularization

- **Intuition**: penalize more complex models to avoid overfitting
- **Regularized ERM approach**: minimize

$$\sum_{i=1}^{N} \ell(f(\mathbf{x}_i; \boldsymbol{\theta}), y_i) + \lambda R(\boldsymbol{\theta})$$

  where $R(\boldsymbol{\theta})$ is a regularizer measuring complexity of $f(\cdot; \boldsymbol{\theta})$
- Given a data set, we control complexity by changing λ
- Common choices: shrinkage regularizers $\|\boldsymbol{\theta}\|_2^2$, $\|\boldsymbol{\theta}\|_1$

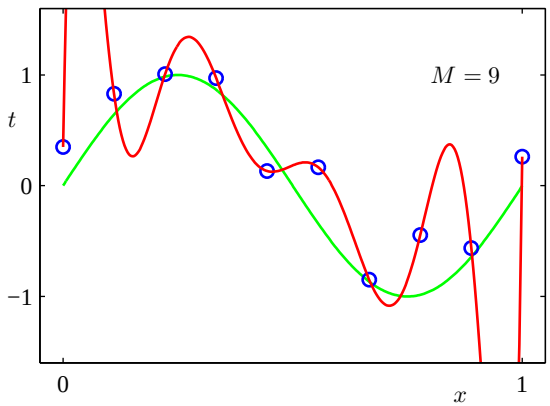
Model selection – tuning hyperparameter(s)

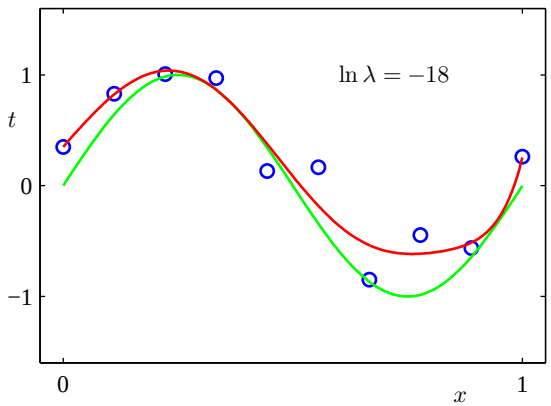
- $\boldsymbol{\theta}$ are model *parameters*; λ is a *hyperparameter*
- Parameters are fit on training data;
  hyperparameters are (typically) tuned on validation/tuning data

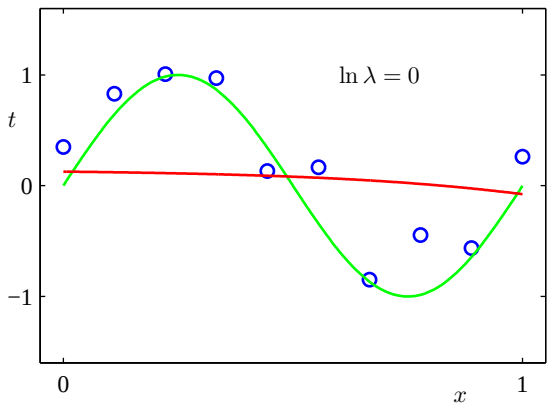
Model selection – tuning hyperparameter(s)

- $\boldsymbol{\theta}$ are model *parameters*; λ is a *hyperparameter*
- Parameters are fit on training data;
  hyperparameters are (typically) tuned on validation/tuning data
- Example [from Bishop, Ch. 1]: 9th deg polynomial with varying λ:

$\lambda = 0$ | $\lambda = e^{-18}$ | $\lambda = 1$







$\|\mathbf{w}^*\|^2 > 10^{12}$ | $\|\mathbf{w}^*\|^2 \approx 21595$ | $\|\mathbf{w}^*\|^2 \approx 0.027$

- Assumption: training, validation, test data are drawn from the true distribution (green), so validation is a proxy for test

Sources of error in learning

- Inherent **uncertainty** (noise) in the data distribution: no way to reduce, other than collect better data (how?)
- **Approximation** error: model class does not contain best possible model; improve by considering a bigger family
- **Estimation** error: model chosen by learning (on finite data) from the class is not the best model in that class; improve by getting more data, simplifying model, regularization
- **Optimization** error: we fail to properly minimize the learning objective, so the learned model is not the best one for the training data; improve via better optimization, different objective (e.g., convex)

Supervised learning II: Classification

Three main "flavors":

- **generative**: estimate $p(y, \mathbf{x}; \boldsymbol{\theta})$
- **discriminative (probabilistic)**: estimate $p(y \mid \mathbf{x}; \boldsymbol{\theta})$
- **discriminative (non-probabilistic)**: directly optimize loss of $f_y(\mathbf{x}; \boldsymbol{\theta})$

Learning classification

- Model computes per-class score $f_y(\mathbf{x}; \boldsymbol{\theta})$
- If perfect separation of training data is impossible, it is intractable to learn a separator minimizing the number of mistakes
- Instead, we learn classifiers by minimizing *surrogate loss* functions

Learning classification

- Model computes per-class score $f_y(\mathbf{x}; \boldsymbol{\theta})$
- If perfect separation of training data is impossible, it is intractable to learn a separator minimizing the number of mistakes
- Instead, we learn classifiers by minimizing *surrogate loss* functions
- Two popular surrogate losses:
  log loss (a.k.a. cross-entropy): assumes $f_y(\mathbf{x}; \boldsymbol{\theta}) \propto \log p\left(y \,|\, \mathbf{x}\right)$

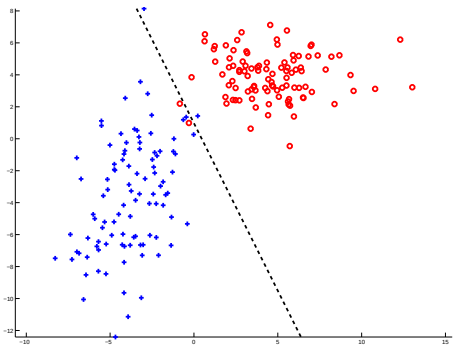
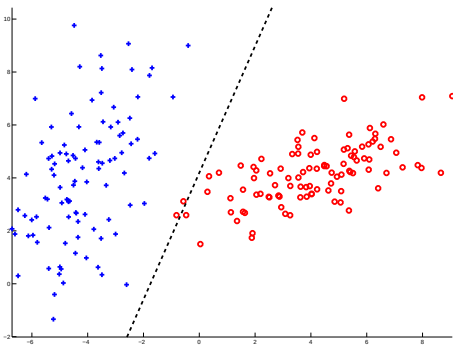
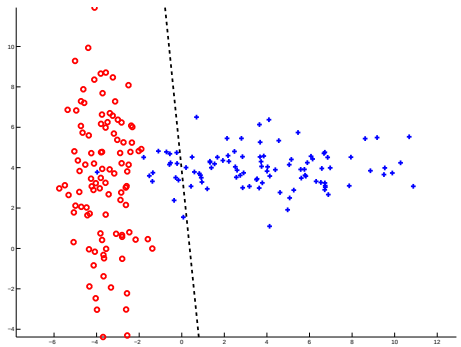
$$\ell(\boldsymbol{\theta}, \mathbf{x}_i, y_i) \;=\; -\log p\left(y_i \,|\, \mathbf{x}_i; \boldsymbol{\theta}\right)$$

  hinge loss (no probabilistic assumption)

$$\ell(\boldsymbol{\theta}, \mathbf{x}_i, y_i) \;=\; \max\left\{0, 1 + \max_{c \neq y_i} f_c(\mathbf{x}_i; \boldsymbol{\theta}) - f_{y_i}(\mathbf{x}_i; \boldsymbol{\theta})\right\}$$

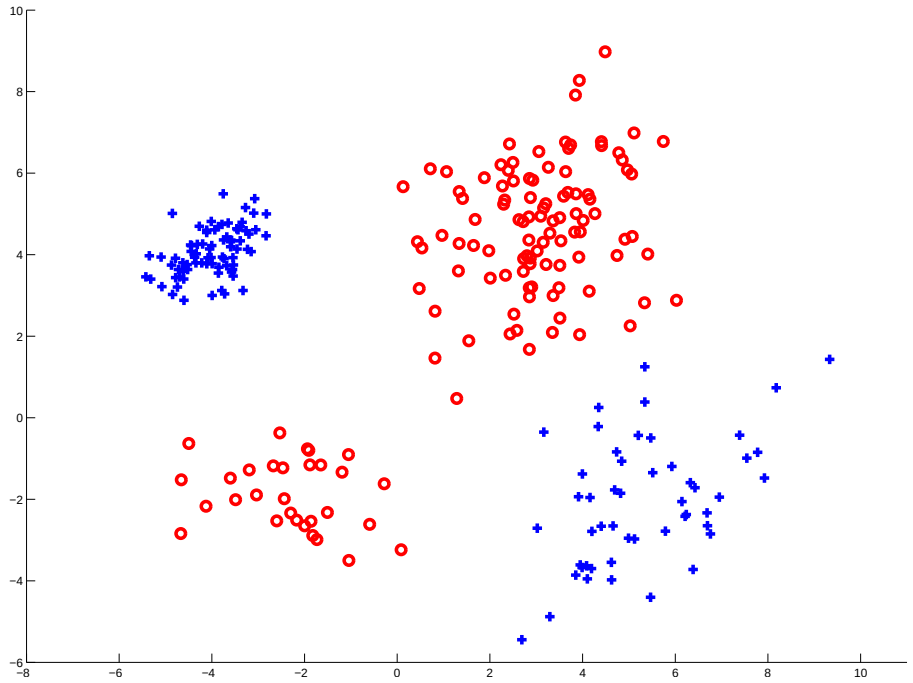
Generative models for classification

- In generative models, most of the "learning" occurs without considering the final classification task
- Typically fit $p(\mathbf{x} \mid y)$ (per-class density) and $p(y)$ (class prior)
- Example: assuming identical-covariance per-class Gaussians yields Linear Discriminant Analysis (LDA); removing the covariance constraint yields Quadratic DA

Density estimation in generative models

- Often we need to go beyond a simple parametric density per class



- Example: mixture models
- Now we are discovering *structure* (patterns) in the training data!
- Generative classification models can be seen as unsupervised learning (density estimation) in the service of a supervised objective

Recap: Supervised learning review

Make sure you are comfortable with key concepts

- Hypothesis classes, loss, risk, empirical risk minimization, task loss vs. surrogate loss
- Regression, classification
- Overfitting, underfitting, regularization, generalization
- Parameters vs. hyperparameters; training, validation/development/tuning, test sets

Other "old" material we will review later

- Kernel methods
- Neural networks

Unsupervised learning setup

- Data set $X = \{\mathbf{x}_i\}$ (no labels)
- Training set, validation set, test set just like in supervised settings
- An objective J defined in terms of $\mathbf{x}_i$
  typically parametric $\{J(X; \boldsymbol{\theta})\}_{\boldsymbol{\theta}}$
- Learning = minimizing the objective over the training data

$$\min_{\boldsymbol{\theta}} \{J(X; \boldsymbol{\theta}) + \lambda R(\boldsymbol{\theta})\}$$

  Note: J may decompose over examples $\mathbf{x}_i$ or not
- As in supervised learning, may overfit with more complex models, so need regularization and model selection (validation set)
- What is J? That is the big question